

VEHICLE PARKING MANAGEMENT SYSTEM

Introduction

The Vehicle Parking Management System is a Python-based project designed to manage vehicles entering and leaving a parking area. The system keeps track of parked vehicles, parking slots, vehicle types and parking charges.

It provides a simple menu-driven interface that makes basic parking management easier and reduces manual record keeping.

Objectives

- To maintain vehicle parking records.
- To assign available parking slots.
- To keep track of parked vehicles.
- To calculate parking charges.
- To manage vehicle entry and exit.
- To reduce manual calculation and record-keeping.

Features

- Park a vehicle
- Assign parking slots
- View parked vehicles
- View available slots
- Remove a vehicle when it exits
- Calculate parking fees
- Display total parking collection
- Simple menu-driven interface

Technology Used

- **Programming Language:** Python
- **Concepts Used:** Variables, Lists, Dictionaries, Functions, Loops and Conditional Statements
- **Development Environment:** Python IDLE / VS Code

PYTHON PROGRAM

```
# Vehicle Parking Management System
```

```
TOTAL_SLOTS = 10
```

```
parking = {}
```

```
total_collection = 0
```

```
def park_vehicle():
```

```
    if len(parking) >= TOTAL_SLOTS:
        print("\nParking is full!")
        return
```

```
    vehicle_number = input("Enter vehicle number: ").upper()
```

```
    vehicle_type = input("Enter vehicle type (Bike/Car): ")
    .capitalize()
```

```
    if vehicle_type not in ["Bike", "Car"]:
        print("Invalid vehicle type!")
        return
```

```
    if vehicle_number in parking:
        print("This vehicle is already parked.")
        return
```

```
    # Find an available slot
```

```
    occupied_slots = [data["slot"] for data in parking.values()]
```

```
    for slot in range(1, TOTAL_SLOTS + 1):
```

```

        if slot not in occupied_slots:
            parking[vehicle_number] = {
                "type": vehicle_type,
                "slot": slot
            }

            print("\nVehicle parked successfully!")
            print("Vehicle Number:", vehicle_number)
            print("Vehicle Type:", vehicle_type)
            print("Parking Slot:", slot)
            return

def view_parked_vehicles():
    if not parking:
        print("\nNo vehicles are currently parked.")
        return

    print("\n==== PARKED VEHICLES =====")

    for vehicle, data in parking.items():
        print(
            "Vehicle:", vehicle,
            "| Type:", data["type"],
            "| Slot:", data["slot"]
        )

def view_available_slots():
    occupied_slots = [data["slot"] for data in parking.values()]

    available_slots = []

    for slot in range(1, TOTAL_SLOTS + 1):
        if slot not in occupied_slots:
            available_slots.append(slot)

    print("\nAvailable Slots:", available_slots)
    print("Number of Available Slots:", len(available_slots))

```

```

def vehicle_exit():
    global total_collection

    vehicle_number = input("Enter vehicle number: ").upper()

    if vehicle_number not in parking:
        print("Vehicle not found in parking.")
        return

    vehicle_type = parking[vehicle_number]["type"]

    print("Vehicle Type:", vehicle_type)

    try:
        hours = int(input("Enter parking duration in hours: "))

        if hours <= 0:
            print("Invalid number of hours.")
            return

    except ValueError:
        print("Please enter a valid number.")
        return

    if vehicle_type == "Bike":
        rate = 20
    else:
        rate = 40

    fee = hours * rate
    total_collection += fee

    slot = parking[vehicle_number]["slot"]

    del parking[vehicle_number]

    print("\n===== PARKING RECEIPT =====")
    print("Vehicle Number:", vehicle_number)
    print("Vehicle Type:", vehicle_type)
    print("Parking Slot:", slot)
    print("Parking Duration:", hours, "hour(s)")

```

```

print("Parking Fee: ₹", fee)
print("Vehicle exited successfully!")

def main():
    while True:
        print("\n=====")
        print(" VEHICLE PARKING MANAGEMENT SYSTEM")
        print("=====")
        print("1. Park Vehicle")
        print("2. View Parked Vehicles")
        print("3. Vehicle Exit")
        print("4. View Available Slots")
        print("5. View Total Collection")
        print("6. Exit")

        choice = input("Enter your choice: ")

        if choice == "1":
            park_vehicle()

        elif choice == "2":
            view_parked_vehicles()

        elif choice == "3":
            vehicle_exit()

        elif choice == "4":
            view_available_slots()

        elif choice == "5":
            print("\nTotal Parking Collection: ₹", total_collection)

        elif choice == "6":
            print("\nThank you for using the Vehicle Parking Management System!")
            break

        else:
            print("\nInvalid choice. Please try again.")

```

```
main()
```

How to Run the Program

1. Install Python on the computer.
2. Open Python IDLE or VS Code.
3. Create a new Python file.
4. Copy the above program into the file.
5. Save the file as:

```
vehicle_parking.py
```

6. Run the program.
7. Select an option from the menu.
8. Enter the required information.

Example Output

```
=====
VEHICLE PARKING MANAGEMENT SYSTEM
=====
1. Park Vehicle
2. View Parked Vehicles
3. Vehicle Exit
4. View Available Slots
5. View Total Collection
6. Exit

Enter your choice: 1

Enter vehicle number: MP04AB1234
Enter vehicle type (Bike/Car): Car

Vehicle parked successfully!
Vehicle Number: MP04AB1234
Vehicle Type: Car
```

Parking Slot: 1

When the vehicle exits:

Enter vehicle number: MP04AB1234

Vehicle Type: Car

Enter parking duration in hours: 3

===== PARKING RECEIPT =====

Vehicle Number: MP04AB1234

Vehicle Type: Car

Parking Slot: 1

Parking Duration: 3 hour(s)

Parking Fee: ₹ 120

Vehicle exited successfully!

Project Files

- `vehicle_parking.py` – Main Python program
- `README.md` – Project description and source code
- `PROJECT_STATEMENT.pdf` – Project statement
- `PROJECT_REPORT.pdf` – Detailed project report
- `VEHICLE_PARKING_SYSTEM.pdf` – System/project documentation

Conclusion

The Vehicle Parking Management System demonstrates how Python can be used to solve a simple real-world parking problem. The program manages vehicle entries, parking slots, vehicle exits and parking fees using basic Python programming concepts.